## Yuchong Pan

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**Contact Information**

MIT, Room 2-341A
77 Massachusetts Avenue
Cambridge, MA 02139
yuchong@mit.edu
ORCID 0000-0002-6134-4232


**Research Interests**

Algorithms, combinatorics, optimization, theoretical computer science, graph theory, operations research.

**Education**

**Massachusetts Institute of Technology**

Ph.D., Mathematics, expected May 2027
- Advisor: Michel X. Goemans

**University of British Columbia**

B.Sc., Computer Science and Mathematics, Honours, with Distinction, Apr 2021
- Thesis: *Optimization Problems on Network Flows with Degree Constraints*
- Advisor: F. Bruce Shepherd

**Employment**

**Microsoft Corporation**

Software Engineer Intern, May 2020–Aug 2020
Software Engineer Intern, Jun 2019–Aug 2019
Software Engineer Intern, May 2018–Aug 2018

**Sogou, Inc.**

Software Engineer Intern, May 2017–Jul 2017

**InitialView**

Software Engineer Intern, Aug 2016–May 2017

**Publications**

- Bicriteria approximation algorithms for demand matching (with M. X. Goemans). In *2027 SIAM Symposium on Simplicity in Algorithms (SOSA)*, 2027, to appear. Preprint available at `https://arxiv.org/abs/2608.04223`.
- Planarity via spanning tree number: a linear-algebraic criterion (with A. Bu). *SIAM J. Discrete Math.*, 39(2), pp. 728–751, 2025. doi:10.1137/24M1660395.

**Manuscripts**

- A better-than-3 approximation algorithm for demand matching via knapsack intersection LP and contention resolution (with M. X. Goemans). *arXiv preprint arXiv:2609.17932*, 2026. URL https://arxiv.org/abs/2609.17932.
- An improved integrality-gap reduction for column-restricted packing integer programs. *Unpublished research note; results obtained in 2026.*
- On high-value and high-flow cycles at basic feasible solutions of subtour elimination relaxations for the symmetric and asymmetric traveling salesman problems (with M. X. Goemans). *Manuscript in preparation; results obtained in 2024.*
- A counterexample to box-half-integrality of the intersection of crossing submodular flow systems (with M. X. Goemans). *Unpublished manuscript*, 2023. URL `https://ypan.me/assets/files/counterexample.pdf`.

- Optimization problems on network flows with degree constraints. Undergraduate honours thesis, University of British Columbia, 2021. URL `https://ypan.me/assets/files/honoursthesis.pdf`.

## Talks and Presentations

- A better-than-3 approximation algorithm for demand matching. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. September 29, 2026.
- Bicriteria approximation algorithms for demand matching. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. September 10, 2026.
- Bicriteria approximation algorithms for demand matching. *The 9th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 21, 2026.
- Planarity via spanning tree number: a linear-algebraic criterion. *The 2025 AMS New England Graduate Student Conference*. Brown University. Providence, RI. April 19, 2025.
- The tales of planar graphs: the extremal problems, the criterion, and the conjectures. *Meet-and-Think*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. February 19, 2025.
- Fantastic planar graphs and where to find them: a linear-algebraic planarity criterion. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. February 10, 2025.
- From submodular flows to planar graphs: a planarity criterion inspired by an OR question. *The 7th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 23, 2024.
- Planarity via spanning tree number: a linear-algebraic criterion. *Algorithms Seminar*. Department of Computer Science, University of British Columbia. Vancouver, BC. May 23, 2024.
- Fantastic spanning trees in planar graphs and where to find them. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. October 12, 2023.
- Perturbation-stable maximum cuts. *Algorithms Reading Group*. Department of Computer Science, University of British Columbia. Online. June 30, 2020.

## Teaching Experience

**Massachusetts Institute of Technology**

*Teaching Assistant (Grader)*

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|---|---|
| 18.435J | Quantum Computation, Fall 2026 |
| | *Same subject as 2.111J, 6.6410J, 8.370J* |
| 18.453/4531 | Combinatorial Optimization, Spring 2023 |
| 18.085/0851 | Computational Science and Engineering I, Fall 2022 |

*Instructor-G (Course Administrator & Recitation Instructor)*

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| 18.06 | Linear Algebra, Spring 2026 |

*Teaching Assistant (Research Mentor)*

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| 18.821 | Project Laboratory in Mathematics, Fall 2025 |

*Teaching Assistant (Recitation Instructor)*

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| 18.600 | Probability and Random Variables, Spring 2025 |
| 18.06 | Linear Algebra, Spring 2024 |

**University of British Columbia**

*Teaching Assistant*

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|---|---|
| CPSC 420 | Advanced Algorithms Design and Analysis, Spring 2021 |
| CPSC 311 | Definition of Programming Languages, Fall 2020 |
| CPSC 421/501 | Introduction to Theory of Computing, Fall 2019 |
| CPSC 121 | Models of Computation, Fall 2018 |

*Academic Assistant*

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| CPSC 411 | Introduction to Compiler Construction, Fall 2019–Spring 2020<br>*Assisted course redesign, supervised by William J. Bowman.* |

## Mentoring Experience

**Massachusetts Institute of Technology**

Directed Reading Program (DRP), Jan 2025–Feb 2025.
- Students: Tiago Oliveira Marques, Eric Shen, Jophy Ye
- Book: *Graphs and Geometry* by László Lovász

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2024–Dec 2024.
- Students: Christopher K. Bao (Davidson Academy, Reno, NV), Joshua T. Wang (Brookfield Central High School, Brookfield, WI), William Zhao (Dougherty Valley High School, San Ramon, CA)
- Topic: Approximate Minimum Cuts in Graphs and Hypergraphs
- A poster titled *A Faster Approximation Algorithm for the k-Way Cut Problem in Low-Rank Hypergraphs via Multiplicative Weight Update* was presented by Christopher K. Bao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Improved Analysis of the Branching Contraction Algorithm for the Unweighted Hypergraph Minimum k-Way Cut Problem* was presented by Joshua T. Wang in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Upper and Lower Bounds on Several Cut Numbers in Hypergraphs via Randomized Contraction* was presented by William Zhao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- Christopher K. Bao, Joshua T. Wang, and William Zhao attend Massachusetts Institute of Technology after high school.

Directed Reading Program (DRP), Jan 2024–Feb 2024.
- Students: Jennifer Ai, Neha Pant
- Book: *Algorithmic Game Theory* edited by N. Nisan, T. Roughgarden, É. Tardos and V. V. Vazirani

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2023–Dec 2023.
- Student: Ruiyang (Raymond) Luo (University High School, Irvine, CA)
- Topic: Cyclic Basis Orderings and Equitability of Matroids
- Ruiyang (Raymond) Luo attends Massachusetts Institute of Technology after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.
- Student: Alan Bu (Phillips Exeter Academy, Exeter, NH)
- Topic: Connections Between Concatenations of Two Incidence Matrices and Spanning Trees in Planar Graphs
- This project was selected as one of the top 5 out of 100 oral presentations.
- Alan Bu won the 10th place (with a $\$40,000$ scholarship) in the 83rd Regeneron Science Talent Search (STS) for this project.
- Alan Bu attends Harvard University after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.
- Student: Deyan Hadzhi-Manich (High School of Mathematics "Dr. Petar Beron", Varna, Bulgaria)
- Topic: Lower and Upper Bounds on Longest Geometrically Increasing Sequences
- Deyan Hadzhi-Manich attends Harvard University after high school.

Directed Reading Program (DRP), Jan 2023–Feb 2023.
- Students: Tianze Jiang, Wayne Zhao
- Book: *Beyond the Worst-Case Analysis of Algorithms* edited by T. Roughgarden

Extended Summer Program in Undergraduate Research (SPUR+), Jun 2022–Aug 2022.
- Students: Evgeniya Artemova, Christina Yu
- Topic: Lower Bounds on the Discrete Newton's Algorithm for the Submodular Line Search Problem

Directed Reading Program (DRP), Jan 2022–Feb 2022.
- Students: Sualeh Asif, Aparna Gupte
- Book: *Randomized Algorithms* by P. Raghavan and R. Motwani

## Honors and Awards

- Graduation with Distinction, University of British Columbia, 2021.
- Science Scholar and Dean's Honour List, University of British Columbia, 2018–2021.
- Scholarships at UBC (total CAD$51,550): Work Learn International Undergraduate Research Award, 2021; Stanley M Grant Scholarship in Mathematics, 2019, 2021; Trek Excellence Scholarship, 2018–2020; Faculty of Science International Student Scholarship, 2018–2020; J Fred Muir Memorial Scholarship in Science, 2020; Dean of Science Scholarship, 2018–2019; Marie Kendall Memorial Scholarship in Science, 2018; Joel Harold Marcoe Memorial Scholarship, 2018; Outstanding International Student Award, 2017.
- 27th Place (out of 118 teams), North American Invitational Programming Contest, Open Division (USA + Canada), 2018.
- 11th Place (out of 67 teams), ACM International Collegiate Programming Contest, Pacific Northwest Regional (Division 1), 2017.
- Silver Medal, IOI China Team Selection Competition, China Computer Federation, 2015.
- Bronze Medal, Asia-Pacific Informatics Olympiad, China Computer Federation, 2015.
- First Prize, National Olympiad in Informatics in Provinces (Grades 10–12 Division), China Computer Federation, 2013–2014.
- Second Prize, National Olympiad in Informatics in Provinces (Grades 7–9 Division), China Computer Federation, 2011–2012.

## Professional Services

Journal Review: SIAM J. Discrete Math., Math. Program., Oper. Res. Lett.
Conference Review: IPCO 2024, SODA 2022

## Relevant Skills

Languages: English, Mandarin
Programming: LaTeX, Python, Julia, C/C++, Java, C#, MATLAB, JavaScript, Go, Racket, Standard ML, MySQL

## Last Updated

October 27, 2026